Applied Economic Forecasting

7. Vector Autoregressive Models (Part II)

1. Impulse Response Functions
2. Forecast Error Variance Decomposition
3. Structural VARs
4. Looking Ahead

Section 1

Impulse Response Functions

From Reduced Form to Dynamics

In Part 1 we estimated the reduced-form VAR and asked: does GDP growth Granger-cause inflation and fed funds rate?

Granger causality tells us **whether** past values of one variable help predict another. It does not tell us:

- **How large** is the effect?
- **In which direction** does it go?
- **How long** does it persist?

Impulse Response Functions (IRFs) answer all three questions.

Definition

An **impulse response function** traces the response of every variable in the system to a one-unit shock in one variable, holding all other shocks at zero, over a horizon of h periods.

The MA(∞) Representation

A stable VAR(p) can be written as a moving average of shocks:

$$\mathbf{y}_t = \boldsymbol{\mu} + \sum_{i=0}^{\infty} \boldsymbol{\Phi}_i \boldsymbol{\varepsilon}_{t-i}$$

where the matrices $\boldsymbol{\Phi}_i$ are the **impulse response matrices** at horizon i.

- $\boldsymbol{\Phi}_0 = \mathbf{I}_K$ (a shock today has an immediate effect)
- $(\boldsymbol{\Phi}_i)_{jk}$ = the response of variable j to a unit shock in variable k, i periods later
- As $i \to \infty$, $\boldsymbol{\Phi}_i \to \mathbf{0}$ for a stable VAR

The identification problem

The reduced-form shocks $\boldsymbol{\varepsilon}_t$ are **correlated** across equations ($\boldsymbol{\Sigma} \neq \mathbf{I}$). We cannot attribute a shock to one variable without contaminating the others. We need to **orthogonalize** the shocks before tracing responses.

Cholesky Identification

The standard approach is a **Cholesky decomposition** of $\mathbf{\Sigma}$:

$$\mathbf{\Sigma} = \mathbf{P}\mathbf{P}'$$

where $\mathbf{P}$ is lower triangular. The orthogonalized shocks are:

$$\mathbf{u}_t = \mathbf{P}^{-1}\boldsymbol{\varepsilon}_t, \quad \mathrm{Var}(\mathbf{u}_t) = \mathbf{I}_K$$

What does the triangular structure impose?

A lower triangular $\mathbf{P}^{-1}$ means the **first variable is affected by no other variable contemporaneously**– it is the most exogenous. The second variable can be affected by the first, but not the third. And so on.

The ordering matters

The Cholesky decomposition imposes a **causal ordering** on the variables. Different orderings produce different IRFs. Important for SVARs.

Our Ordering

Recall our system:

$$\mathbf{y}_t = \begin{pmatrix} g_t \\ \pi_t \\ r_t \end{pmatrix} = \begin{pmatrix} \text{GDP growth} \\ \text{Inflation} \\ \Delta\text{Fed Funds Rate} \end{pmatrix}$$

The ordering imposes:

- GDP growth does not respond contemporaneously to inflation or monetary policy shocks (slow-moving real economy)
- Inflation responds to GDP growth within the quarter, but not to monetary policy
- The fed funds rate can respond to both GDP growth and inflation within the quarter (fast-moving policy instrument)

Is this reasonable?

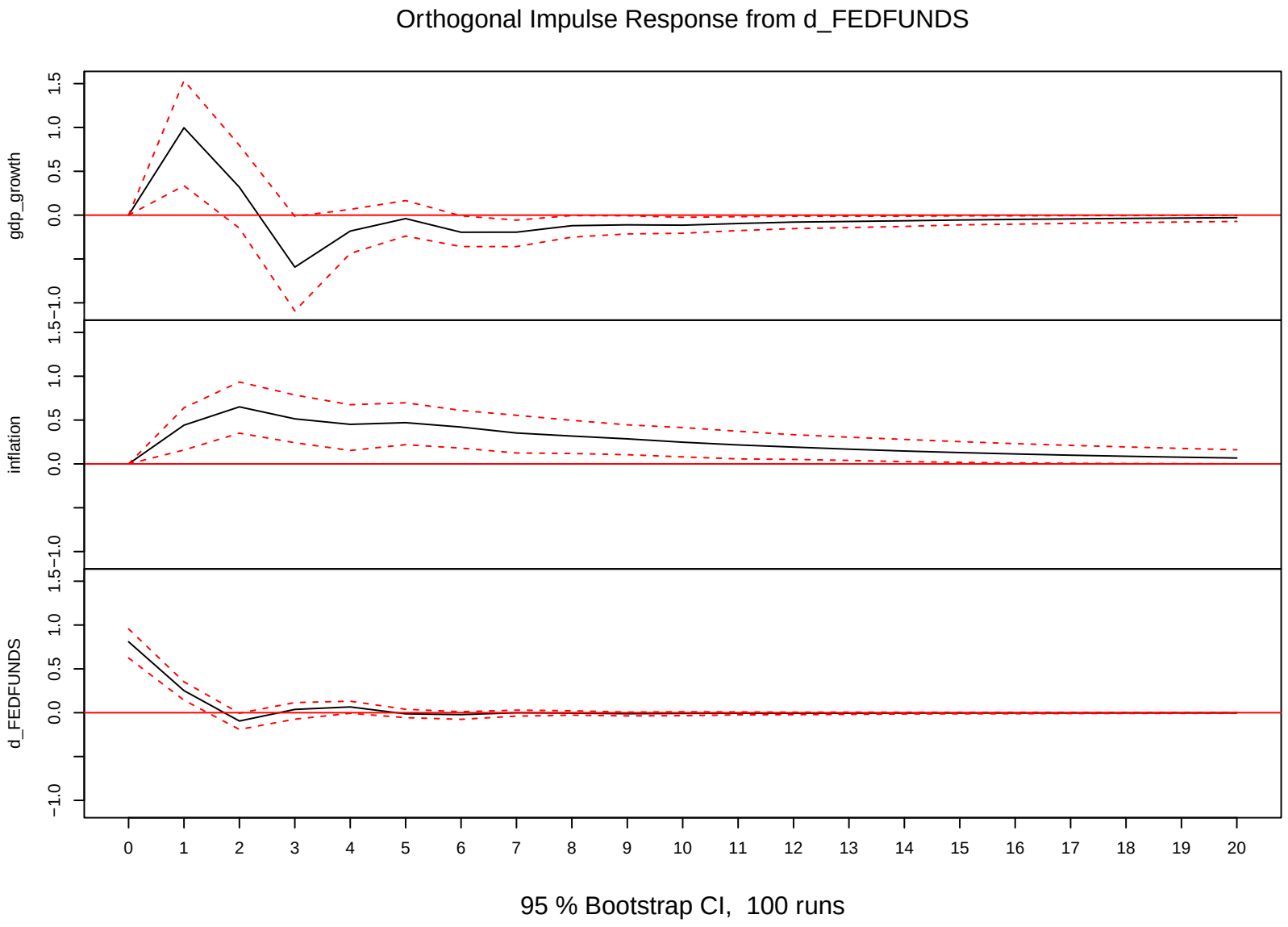
Standard recursive ordering in the monetary VAR literature (Christiano, Eichenbaum, Evans 1999). It embeds a simple Taylor-rule timing

IRFs in R

```r
## Orthogonalized IRFs -- Cholesky identification
## Response of all variables to a shock in d_FEDFUNDS

irf.ff <- vars::irf(var.fit3,
                    impulse  = "d_FEDFUNDS",
                    response = c("gdp_growth",
                                 "inflation",
                                 "d_FEDFUNDS"),
                    n.ahead  = 20,
                    ortho    = TRUE,
                    boot     = TRUE,
                    ci       = 0.95)

plot(irf.ff)
```


Orthogonal Impulse Response from d_FEDFUNDS
gdp_growth
inflation
d_FEDFUNDS
95 % Bootstrap CI, 100 runs

Interpreting the IRF: Fed Funds Rate Shock

What are we looking at?

- The horizontal axis is the horizon (quarters after the shock)
- The vertical axis is the response in the units of that variable (% for growth rates, percentage points for the rate)
- The solid line is the point estimate; the shaded band is the 95% bootstrap confidence interval
- A response is **statistically significant** at horizon h if the confidence band excludes zero

Interpreting the IRF: Fed Funds Rate Shock

Economic interpretation:

- A tightening shock (+1 unit in Δr_t) initially produces a **positive** response in GDP growth– peaking near horizon 1 before turning negative around quarters 2–3
- The response of inflation is also **positive and persistent** — the confidence band excludes zero for much of the horizon

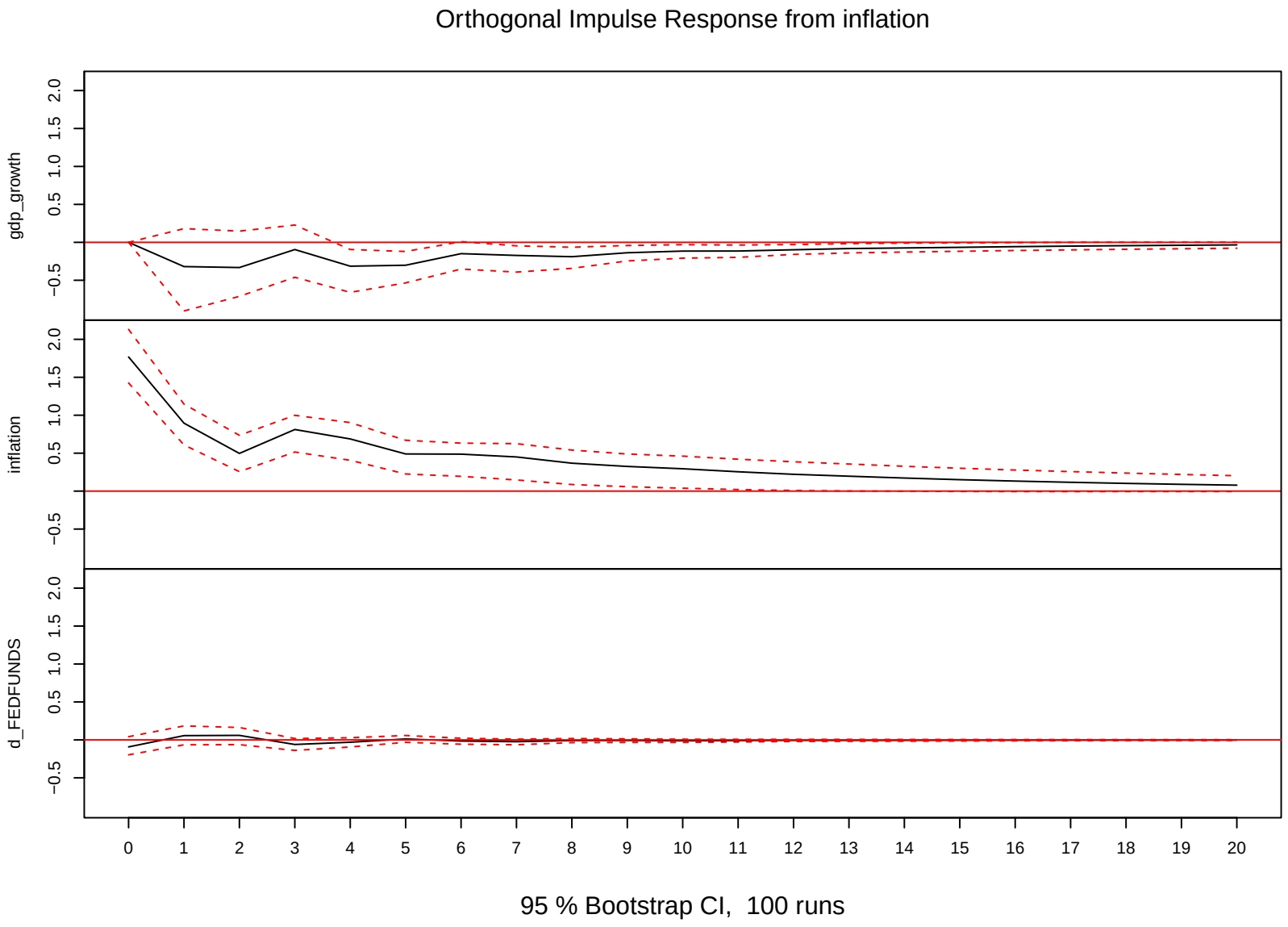
The price puzzle

The positive inflation response to a tightening shock is the well-known "price puzzle" (Sims 1992). Rather than falling, inflation rises after a contractionary monetary shock in recursive VARs. Two leading explanations:

1. The fed funds rate is ordered last — it cannot respond contemporaneously to inflation, so the model cannot distinguish a policy tightening from an endogenous reaction to rising inflation
2. The system omits forward-looking variables (e.g., commodity prices) that the Fed monitors but that are not in our three-variable system

IRF: Response to an Inflation Shock

```r
irf.inf <- vars::irf(var.fit3,
                     impulse  = "inflation",
                     response = c("gdp_growth",
                                  "inflation",
                                  "d_FEDFUNDS"),
                     n.ahead  = 20,
                     ortho    = TRUE,
                     boot     = TRUE,
                     ci       = 0.95)
plot(irf.inf)
```


Orthogonal Impulse Response from inflation
gdp_growth
inflation
d_FEDFUNDS
2.0
1.5
1.0
0.5
−0.5
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
95 % Bootstrap CI, 100 runs

Interpreting the IRF: Inflation Shock

What the output shows:

- Inflation responds strongly and persistently to its own shock — the confidence band excludes zero throughout, confirming high inflation persistence in this sample
- GDP growth responds **negatively** but the effect is small and the confidence band spans zero at most horizons — not statistically significant
- The fed funds rate response is **near zero and insignificant** — the expected monetary policy tightening in response to inflation does not appear

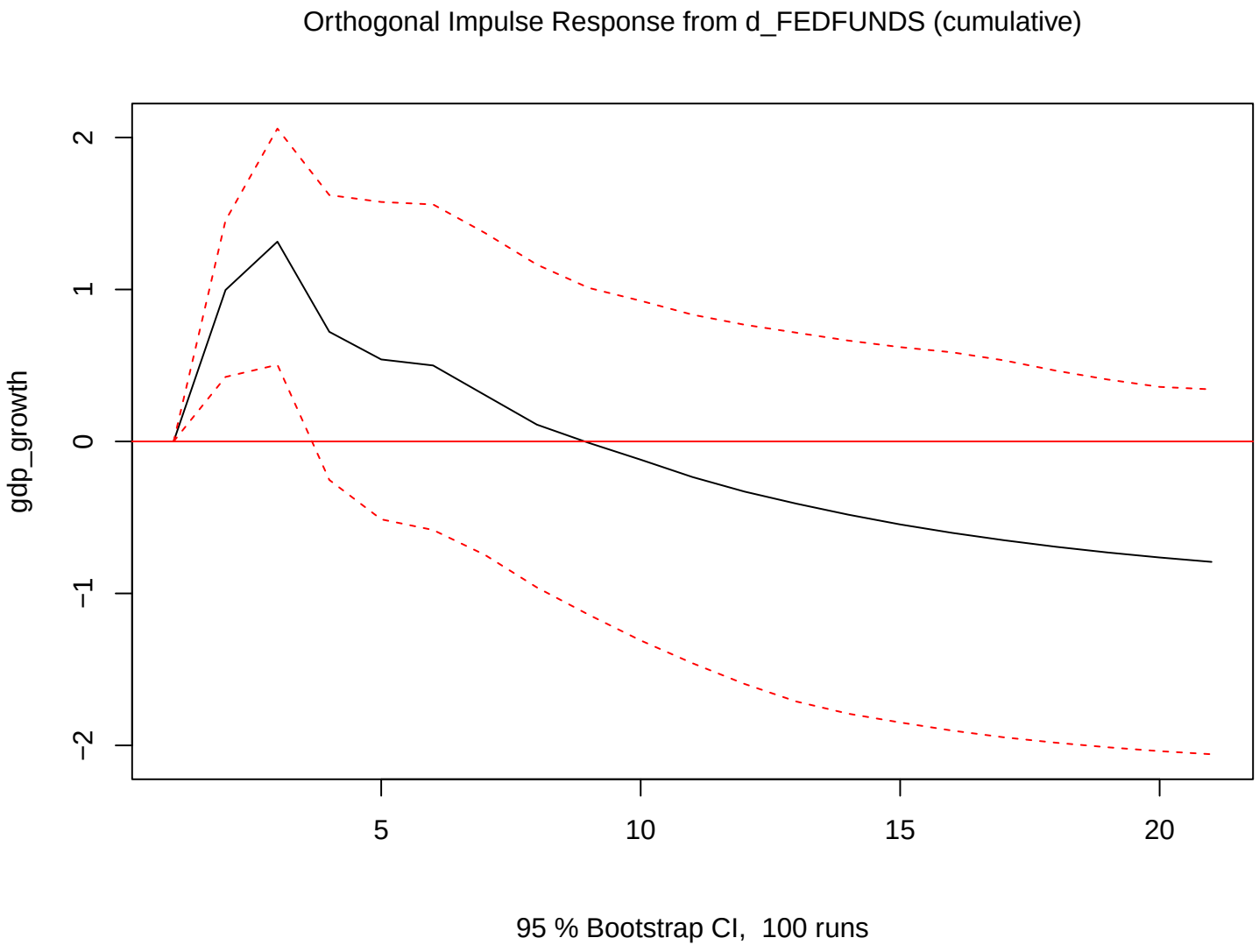
Connecting back to Granger causality

Granger causality told us inflation has predictive content for the system ($p = 0.03$). The IRF shows us where that content lives: primarily in inflation's own persistence. The expected Taylor rule response– the Fed raising rates in response to an inflation shock– is not apparent here, likely reflecting the same ordering limitation as the price puzzle. This

Cumulative IRFs

```r
## Cumulative IRF -- useful for level effects
## (e.g., effect on the level of GDP, not just its
## growth rate)

irf.cum <- vars::irf(
  var.fit3,
  impulse    = "d_FEDFUNDS",
  response   = "gdp_growth",
  n.ahead    = 20,
  ortho      = TRUE,
  cumulative = TRUE,
  boot       = TRUE,
  ci         = 0.95)
plot(irf.cum)
```


Orthogonal Impulse Response from d_FEDFUNDS (cumulative)
gdp_growth
2
1
0
−1
−2
5
10
15
20
95 % Bootstrap CI, 100 runs

Cumulative IRFs

When are cumulative IRFs useful?

- Our system models GDP *growth*, not the level of GDP
- A shock that depresses growth for several quarters compounds into a level effect
- The cumulative IRF integrates the growth response across horizons, recovering the implied level effect

What the output shows

The cumulative response of GDP growth to a fed funds shock is initially **positive** (reflecting the impact-period GDP growth increase), crosses zero around quarters 7–8, and turns negative at longer horizons.
However, the 95% bootstrap confidence band spans zero from roughly horizon 4 onward — the cumulative level effect is **not statistically significant** at conventional levels.
This is consistent with the price puzzle finding: the tightening shock in this reduced-form system does not cleanly identify a contractionary monetary policy effect. Structural identification — which we turn to in

Section 2

Forecast Error Variance Decomposition

What Drives Forecast Uncertainty?

When we produce a multi-step forecast from a VAR, the forecast error grows with the horizon. But *where does that uncertainty come from*?

The **Forecast Error Variance Decomposition (FEVD)** answers:

Definition

The FEVD decomposes the h-step-ahead forecast error variance of variable j into the shares attributable to shocks in each variable $k = 1, \ldots, K$.

At horizon h, the share of the forecast error variance of y_j attributable to shocks in y_k:

$$\omega_{jk}(h) = \frac{\sum_{i=0}^{h-1} (\mathbf{\Phi}_i \mathbf{P})_{jk}^2}{\sum_{i=0}^{h-1} \sum_{\ell=1}^{K} (\mathbf{\Phi}_i \mathbf{P})_{j\ell}^2}$$

where $\sum_{k=1}^{K} \omega_{jk}(h) = 1$ at every horizon.

FEVD: Interpretation

What does the FEVD tell us?

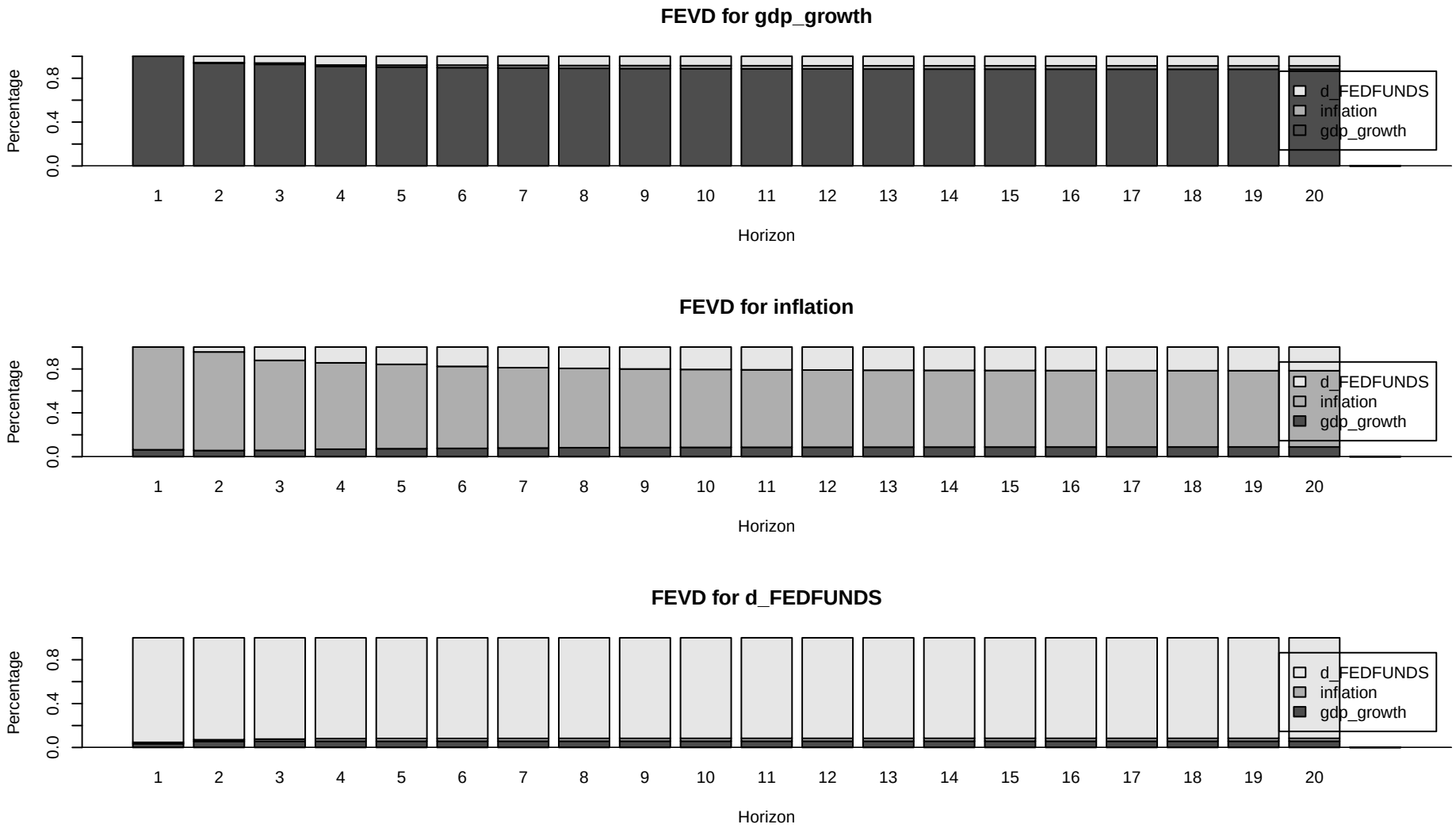
- At **short horizons**, own shocks dominate– each variable is mostly driven by its own innovations
- As the horizon grows, shocks from other variables become more important as they propagate through the system
- A large share from variable k at long horizons suggests k is an important *structural driver* of j's variation– not just a short-run predictor

Difference from Granger causality:

Granger causality is a binary test (does y_2 help predict y_1?). FEVD is quantitative– it tells us *how much* of y_1's forecast uncertainty is due to shocks in y_2 at each horizon. A variable can Granger-cause another while accounting for only a small share of its variance.

FEVD in R

```r
fevd.fit <- vars::fevd(var.fit3, n.ahead = 20)
plot(fevd.fit)
```



FEVD: Tabular Summary

```r
make_fevd_table <- function(fevd_element,
      h = c(1, 4, 8, 12, 20),
      row_names = c("GDP Growth", "Inflation", "Fed Funds Rate")) {

  (fevd_element %>%
     data.frame())[h, ] %>% t() %>%
    # Name the rows
    `rownames<-`(row_names) %>%
    knitr::kable(digits = 3) %>%
    # Merge and center Horizon columns
    kableExtra::add_header_above(c(" " = 1, "Horizon" = length(h)))
}
```

```r
make_fevd_table(fevd.fit$gdp_growth)
```

	Horizon				
	1	4	8	12	20
GDP Growth	1	0.908	0.891	0.885	0.882
Inflation	0	0.012	0.025	0.029	0.031
Fed Funds Rate	0	0.080	0.084	0.086	0.087

```r
make_fevd_table(fevd.fit$inflation)
```

	Horizon				
	1	4	8	12	20
GDP Growth	0.063	0.069	0.082	0.087	0.089
Inflation	0.937	0.788	0.723	0.704	0.695
Fed Funds Rate	0.000	0.144	0.194	0.210	0.216

```r
make_fevd_table(fevd.fit$d_FEDFUNDS)
```

	Horizon				
	1	4	8	12	20
GDP Growth	0.034	0.057	0.057	0.057	0.057
Inflation	0.012	0.024	0.026	0.026	0.026
Fed Funds Rate	0.953	0.919	0.917	0.916	0.916

FEVD: Economic Interpretation (Generic)

What to look for:

- **GDP growth**: What share of its long-run forecast uncertainty is attributable to monetary policy shocks versus own shocks? A large fed funds share suggests monetary policy is an important driver of business cycle fluctuations.
- **Inflation**: Does the fed funds rate account for a meaningful share of inflation variance at longer horizons? This speaks to the effectiveness of monetary policy in anchoring inflation.
- **Fed funds rate**: Is most of its variance explained by own shocks (autonomous policy) or by inflation and output (systematic reaction function)?

FEVD: Economic Interpretation

What the numbers tell us:

- **GDP growth** is driven almost entirely by its own shocks ($\approx$ 88–89% at all horizons). The fed funds rate accounts for only $\approx$ 8% even at 20 quarters: monetary policy shocks explain little of output variance in this system.
- **Inflation** shows the most interesting dynamics. Own shocks fall from 94% at horizon 1 to 70% at horizon 20, while the fed funds rate share *rises* from 0% to 22%. Monetary policy becomes an increasingly important driver of inflation variance at longer horizons.
- **Fed funds rate** is overwhelmingly driven by its own shocks ($\approx$ 92–95%). Inflation and output together account for less than 10%: the policy rate behaves largely autonomously in this system.

FEVD: Economic Interpretation

Connecting to the price puzzle

The small fed funds share in GDP growth variance, and the delayed but growing share in inflation variance, are consistent with a system where the monetary transmission mechanism is weak or mis-identified. The Cholesky ordering places the fed funds rate last, limiting its contemporaneous influence. These findings reinforce the motivation for structural identification.

FEVD: Economic Interpretation

A general finding in the macro VAR literature

Own shocks typically dominate at short horizons. At horizons beyond 8 quarters, monetary policy shocks often account for 10–25% of output variance and 5–15% of inflation variance, though results vary substantially across samples and specifications.

Section 3

Structural VARs

The Identification Problem

The reduced-form VAR we have estimated is:

$$\mathbf{y}_t = \mathbf{c} + \mathbf{A}_1\mathbf{y}_{t-1} + \cdots + \mathbf{A}_p\mathbf{y}_{t-p} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim (0, \boldsymbol{\Sigma})$$

The reduced-form shocks $\boldsymbol{\varepsilon}_t$ are **not structural** — they are linear combinations of the underlying economic shocks (supply shocks, demand shocks, policy shocks, etc.).

Why does this matter?

- We cannot say: "this is a monetary policy shock''
- The IRFs we computed earlier are valid forecasting tools, but their economic interpretation depends on the identification assumption
- The Cholesky decomposition we used imposes a *particular* set of restrictions– and those restrictions may not be economically justifiable

The Structural VAR

A **Structural VAR (SVAR)** imposes restrictions on the contemporaneous relationships among variables to recover economically meaningful shocks.

The structural form is written as:

$$\mathbf{A}\boldsymbol{\varepsilon}_t = \mathbf{u}_t, \qquad \mathbf{u}_t \sim (0, \mathbf{I}_K)$$

where $\mathbf{A}$ captures the **contemporaneous structural relationships** among the reduced-form shocks, and $\mathbf{u}_t$ are the orthogonal structural shocks.

- $\mathbf{A}$ is the matrix we restrict. Zeros in $\mathbf{A}$ impose that certain variables do not respond contemporaneously to certain shocks
- The reduced-form errors are recovered as $\boldsymbol{\varepsilon}_t = \mathbf{A}^{-1}\mathbf{u}_t$

The Structural VAR

The link to the reduced form:

$$\boldsymbol{\varepsilon}_t = \mathbf{A}^{-1}\mathbf{u}_t, \qquad \boldsymbol{\Sigma} = \mathbf{A}^{-1}(\mathbf{A}^{-1})'$$

The identification problem, precisely

$\boldsymbol{\Sigma}$ has $K(K+1)/2$ free parameters. $\mathbf{A}$ has K^2.

We need at least $K(K-1)/2$ restrictions on $\mathbf{A}$ to identify the system. With $K = 3$: at least **3 restrictions** are needed.

In `vars::SVAR()`:

- `Amat` imposes restrictions on $\mathbf{A}$ directly. `NA` means the element is free to be estimated, `0` imposes a zero restriction.
- The short-run (Cholesky) identification sets $\mathbf{A}$ to be lower triangular with free diagonal (exactly 3 zero restrictions for $K = 3$)

Where the Reduced-Form VAR Falls Short

The reduced-form VAR is perfectly adequate for:

- **Forecasting:** no identification needed; we just need a good statistical model of the joint distribution of $\mathbf{y}_t$
- **Granger causality:** asessing predictive content
- **Reduced-form IRFs:** quantifying dynamic correlations

But it is not sufficient for:

- **Structural impulse responses:** "what happens to output after a monetary policy tightening?" requires knowing which shock is the policy shock
- **Policy counterfactuals:** what would inflation have been without the 2022 rate hikes?
- **Testing structural hypotheses:** does the Fed respond more to inflation than to output?

The SVAR is the tool when economic interpretation of shocks matters.

Identification Strategies

Three common approaches:

1. Short-run restrictions (Cholesky / recursive)

Impose zeros on $\mathbf{A}$: some variables do not respond contemporaneously to certain shocks– think back to the Cholesky decomposition. Easy to implement; requires a defensible causal ordering.

2. Long-run restrictions (Blanchard-Quah)

Impose restrictions on the **cumulative** effect of shocks. For example: demand shocks have no long-run effect on the level of output (only supply shocks do). Requires economic theory about long-run neutrality.

3. Sign restrictions

Rather than exact zeros, impose that certain shocks move certain variables in specified *directions* (e.g., a monetary tightening shock raises rates *and* lowers inflation). More agnostic about the exact contemporaneous structure; implemented via Bayesian methods.

SVAR in R: Short-Run Restrictions

```r
## Short-run (Cholesky) SVAR
## A-model: lower triangular identification
## Diagonal = identity (scaling); upper triangle = 0 (restricted)
## Lower triangle = NA (free to estimate)
k <- 3
amat <- diag(k)
#amat[1, 2] = 0, GDP growth not affected by inf
#amat[1, 3] = 0, GDP growth not affected by FFR shock
#amat[2, 3] = 0, inf. not affected by FFR shock
amat[lower.tri(amat)] <- NA #Add. coefs to estimate

svar.sr <- vars::SVAR(var.fit3,
                      estmethod = "scoring",
                      Amat = amat)
summary(svar.sr)
```

Note

vars::SVAR() accepts either an **Amat** (contemporaneous coefficient restrictions) or a **Bmat** (shock covariance restrictions), depending on the parameterization. The short-run recursive model above is equivalent to the Cholesky IRF we computed earlier– it is a useful starting point before imposing more substantive restrictions.

Structural vs. Reduced-Form IRFs

Once the SVAR is identified, structural IRFs are computed in exactly the same way– i.e. replacing the Cholesky $\mathbf{P}$ with the estimated $\hat{\mathbf{B}}_0^{-1}$.

In R:

```
## Structural IRF from the SVAR
irf.svar <- vars::irf(svar.sr,
                      impulse  = "d_FEDFUNDS",
                      response = c("gdp_growth",
                                   "inflation",
                                   "d_FEDFUNDS"),
                      n.ahead  = 20,
                      boot     = TRUE)
plot(irf.svar)
```

Structural vs. Reduced-Form IRFs

Comparing reduced-form and structural IRFs

If the short-run SVAR uses the same Cholesky ordering as our reduced-form IRFs, the results will be numerically identical. The SVAR framework becomes important when you want to impose *different* or *over-identifying* restrictions that go beyond what Cholesky allows.

When to Use a VAR vs. an SVAR

Task	**VAR**	**SVAR**
Forecasting	✓	✓
Granger causality	✓	
Reduced-form IRFs	✓	
Structural shock identification		✓
Policy counterfactuals		✓
Testing structural hypotheses		✓

The practical takeaway

1. Start with the reduced-form VAR.
2. If your goal is forecasting or descriptive dynamics, you may not need to go further.
3. If you want to make causal statements about structural shocks (e.g. what is the effect of a monetary policy surprise?) the SVAR is the right tool. Choice of identification scheme is where the economic content lives.

Section 4

Looking Ahead

Part 2 Summary

What we covered:

1. **Impulse Response Functions:** tracing dynamic responses to orthogonalized shocks; Cholesky identification; ordering rationale
2. **Cumulative IRFs:** recovering level effects from growth-rate systems
3. **Forecast Error Variance Decomposition:** quantifying the share of forecast uncertainty attributable to each structural driver
4. **Structural VARs:** the identification problem; short-run, long-run, and sign restrictions; when the SVAR is needed

Part 3: Other Considerations?? Cointegration, the Johansen test, and Vector Error Correction Models (VECMs) — the right framework when variables share a long-run equilibrium relationship.